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Automatic lead identification using electric field fingerprinting

Abstract

A method, programmer for a neurostimulator, and neurostimulation kit are provided. The kit comprises a neurostimulator, and a plurality of elongated lead bodies configured for being coupled to the neurostimulator, each having a plurality of proximal contacts and a plurality of distal electrodes respectively electrically coupled to the proximal contacts, wherein an in-line connectivity between the electrodes and proximal contacts carried by the different lead bodies differs from each other. Electrical energy is conveyed between the electrodes of the selected lead body and the tissue, an electrical fingerprint is measured at the proximal contacts of the selected lead body in response to the conveyed electrical energy, and the selected lead body is identified based on the measured electrical fingerprint. These steps can be performed by the programmer.

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Background/Summary

RELATED APPLICATION (1) The present application is a continuation of U.S. application Ser. No. 17/752,581, filed May 24, 2022, which is a continuation of U.S. application Ser. No. 16/719,793, filed Dec. 18, 2019, which is a continuation of U.S. application Ser. No. 16/050,546, filed Jul. 31, 2018, now issued as U.S. Pat. No. 10,549,085, which is a continuation of U.S. application Ser. No. 15/674,769, filed Aug. 11, 2017, now issued as U.S. Pat. No. 10,052,475, which is a continuation of U.S. application Ser. No. 12/941,657, filed Nov. 8, 2010, now issued as U.S. Pat. No. 9,731,106, which claims the benefit under 35 U.S.C. § 119 to U.S. provisional patent application Ser. No. 61/259,509, filed Nov. 9, 2009. The foregoing applications are hereby incorporated by reference into the present application in their entirety.

FIELD OF THE INVENTION

(1) The present invention relates to tissue stimulation systems, and more particularly, to apparatus and methods for identifying neurostimulation leads.

BACKGROUND OF THE INVENTION

(2) Implantable neurostimulation systems have proven therapeutic in a wide variety of diseases and disorders. Pacemakers and Implantable Cardiac Defibrillators (ICDs) have proven highly effective in the treatment of a number of cardiac conditions (e.g., arrhythmias). Spinal Cord Stimulation (SCS) systems have long been accepted as a therapeutic modality for the treatment of chronic pain syndromes, and the application of tissue stimulation has begun to expand to additional applications such as angina pectoralis and incontinence. Deep Brain Stimulation (DBS) has also been applied therapeutically for well over a decade for the treatment of refractory chronic pain syndromes, and DBS has also recently been applied in additional areas such as movement disorders and epilepsy. Further, in recent investigations Peripheral Nerve Stimulation (PNS) systems have demonstrated efficacy in the treatment of chronic pain syndromes and incontinence, and a number of additional applications are currently under investigation. Also, Functional Electrical Stimulation (FES) systems such as the Freehand system by NeuroControl (Cleveland, Ohio) have been applied to restore some functionality to paralyzed extremities in spinal cord injury patients.

(3) Each of these implantable neurostimulation systems typically includes one or more stimulation leads implanted at the desired stimulation site and an implantable neurostimulator, such as an implantable pulse generator (IPG), implanted remotely from the stimulation site, but coupled either directly to the stimulation leads or indirectly to the stimulation leads via one or more lead extensions in cases where the length of the stimulation leads is insufficient to reach the IPG. Thus, electrical pulses can be delivered from the neurostimulator to the stimulation leads to stimulate the

tissue and provide the desired efficacious therapy to the patient.

(4) In the context of an SCS procedure, one or more stimulation leads are introduced through the patient's back into the epidural space under fluoroscopy, such that the electrodes carried by the leads are arranged in a desired pattern and spacing to create an electrode array. The specific procedure used to implant the stimulation leads will ultimately depend on the type of stimulation leads used. Currently, there are two types of commercially available stimulation leads: a percutaneous lead and a surgical lead.

(5) A percutaneous lead comprises a cylindrical body with ring electrodes, and can be introduced into contact with the affected spinal tissue through a Touhy-like needle, which passes through the skin, between the desired vertebrae, and into the epidural space above the dura layer. For unilateral pain, a percutaneous lead is placed on the corresponding lateral side of the spinal cord. For bilateral pain, a percutaneous lead is placed down the midline of the spinal cord, or two percutaneous leads are placed down the respective sides of the midline.

(6) A surgical lead has a paddle on which multiple electrodes are arranged in independent columns, and is introduced into contact with the affected spinal tissue using a surgical procedure, and specifically, a laminectomy, which involves removal of the laminar vertebral tissue to allow both access to the dura layer and positioning of the lead.

(7) After proper placement of the stimulation leads at the target area of the spinal cord, the leads are anchored in place at an exit site to prevent movement of the stimulation leads. To facilitate the location of the neurostimulator away from the exit point of the stimulation leads, lead extensions are sometimes used. Whether lead extensions are used or not, the proximal ends of the stimulation leads exiting the spinal column are passed through a tunnel subcutaneously formed along the torso of the patient to a subcutaneous pocket (typically made in the patient's abdominal or buttock area) where a neurostimulator is implanted. The subcutaneous tunnel can be formed using a tunneling tool over which a tunneling straw may be threaded. The tunneling tool can be removed, the stimulation leads threaded through the tunneling straw, and then the tunneling straw removed from the tunnel while maintaining the stimulation leads in place within the tunnel.

(8) The stimulation leads are then connected directly to the neurostimulator by inserting the proximal ends of the stimulation leads within one or more connector ports of the IPG or connected to lead extensions, which are then inserted into the connector ports of the IPG. The IPG can then be operated to generate electrical pulses that are delivered, through the electrodes, to the targeted tissue, and in particular, the dorsal column and dorsal root fibers within the spinal cord.

(9) The stimulation creates the sensation known as paresthesia, which can be characterized as an alternative sensation that replaces the pain signals sensed by the patient. Intra-operatively (i.e., during the surgical procedure), the neurostimulator may be operated to test the effect of stimulation and adjust the parameters of the stimulation for optimal pain relief. The patient may provide verbal feedback regarding the presence of paresthesia over the pain area, and based on this feedback, the lead positions may be adjusted and re-anchored if necessary. A computerized programming system, such as Bionic Navigator®, available from Boston Scientific Corporation, can be used to facilitate selection of the stimulation parameters. Any incisions are then closed to fully implant the system. Post-operatively (i.e., after the surgical procedure has been completed), a clinician can adjust the stimulation parameters using the computerized programming system to re-optimize the therapy.

(10) Oftentimes, multiple lead bodies may extend from the spinal region of the patient. For example, multiple percutaneous leads may be implanted within the patient adjacent the spinal cord, or in the case of paddle leads, multiple lead tails may extend from the paddle, with each lead tail being coupled to specific electrodes on the paddle. Because the programming of the IPG will depend upon the physical locations of the electrodes relative to the patient's spinal cord, the proximal ends of the lead bodies are labeled before passing them through the tunneling straw, so that the surgeon can keep track of which set of electrodes is connected to which connector port on the implanted IPG (which may include up to four ports in the near future), or if multiple IPGs are

to be implanted, which set of electrodes is connected to which IPG.

(11) One technique used by surgeons to identify the lead bodies is to tie sutures around the proximal ends of the lead bodies prior to introducing them through the tunneling straw; for example, one suture around a first lead body, two sutures around a second lead body, three sutures around a third lead body, etc. Once the proximal ends of the lead bodies exit the tunneling straw, the surgeon can then identify each lead body (and thus the corresponding electrodes) by the number of sutures tied to the respective lead body, thereby allowing the lead body to be connected to the correct port on the IPG.

(12) While this technique can be successfully employed to identify lead bodies, it considerably extends the length of the surgery time, which is undesirable. In some cases, the identification features, such as different colors or markings, can be incorporated into the proximal ends of the lead bodies, such that the lead bodies can be identified as they exit the tunneling straw. Even with the use of visual identifiers, however, the proximal ends of the lead bodies can still be inserted into the incorrect connector ports. If the lead bodies are inserted into the incorrect connector ports, intra-operative testing of the lead placement may be compromised. Additional surgical time may be wasted to identify and correct the connection problem. If the errors remain unidentified, the patient may leave the operating room with the lead bodies incorrectly connected. During post-operative fitting, additional time may then be lost identifying and compensating for lead bodies that are not in the proper connector ports. This ultimately can result in sub-optimal therapy.

(13) Another related problem arises when different types of stimulation leads are used. During an intra-operative or post-operative procedure, the clinician is required to select the model of the stimulation lead or leads in order for the computerized programming system to operate in an optimal fashion. As the number of lead models increases, it becomes increasingly likely that the incorrect lead model will be selected.

(14) There, thus, remains a need for a quick, effective, and low-cost method for identifying a lead body of a neurostimulation lead.

SUMMARY OF THE INVENTION

(15) In accordance with a first aspect of the present inventions, a method of identifying a selected one of a plurality of different lead bodies is provided. The plurality of lead bodies may comprises, e.g., at least two lead bodies of a surgical paddle lead and/or a lead body of a surgical paddle lead and a lead body of a percutaneous lead. Each of the lead bodies includes a plurality of proximal contacts and a plurality of distal electrodes respectively electrically coupled to the proximal contacts. The selected lead body is in contact with tissue of a patient.

(16) The method comprises conveying electrical energy between the electrodes of the selected lead body and the tissue. The method further comprises measuring an electrical fingerprint at the proximal contacts of the selected lead body in response to the conveyed electrical energy. By way of non-limiting example, the measured electrical fingerprint may comprise a plurality of electrical field potentials respectively measured at the proximal contacts of the selected lead body. The method further comprises identifying the selected lead body based on the measured electrical fingerprint. In one method, the in-line connectivity between the electrodes and proximal contacts carried by the different lead bodies differs from each other. In this manner, the measured electrical fingerprints will differ from each other.

(17) By way of non-limiting example, the method may further comprise comparing the measured electrical fingerprint to each of a plurality of reference electrical fingerprints corresponding to the different lead bodies, determining a match between the measured electrical fingerprint and one of the reference electrical fingerprints, and identifying the selected lead body as the lead body corresponding to the matching one of the reference electrical fingerprints. The data points in the measured electrical fingerprint may be computationally compared to data points in each of the reference electrical fingerprints. For example, the computational comparison for each of the reference electrical fingerprints may comprise generating a correlation coefficient (e.g., a Pearson

correlation coefficient), wherein a match between the measured electrical fingerprint and the one reference electrical fingerprint is determined based on the values of the generated correlation coefficients.

(18) In accordance with a second aspect of the present inventions, a programmer for a neurostimulator is provided. The programmer comprises a user interface configured for receiving user commands, and input circuitry configured for receiving an electrical fingerprint measured by a neurostimulator connected to a lead body. The programmer further comprises a processor configured for automatically identifying the lead body from a plurality of different lead bodies based on the measured electrical fingerprint, and generating stimulation parameters for the identified lead body in response to the user commands. The plurality of lead bodies may comprises, e.g., at least two lead bodies of a surgical paddle lead and/or a lead body of a surgical paddle lead and a lead body of a percutaneous lead. By way of non-limiting example, the measured electrical fingerprint may comprise a plurality of electrical field potentials respectively measured at the proximal contacts of the selected lead body. The programmer further comprises output circuitry (e.g., telemetry circuitry) for transmitting the stimulation parameters to the neurostimulator.

(19) In one embodiment, the programmer further comprises memory storing a plurality of reference electrical fingerprints corresponding to the different lead bodies. In this case, the processor is configured for comparing the measured electrical fingerprint to each of a plurality of reference electrical fingerprints, determining a match between the measured electrical fingerprint and one of the reference electrical fingerprints (e.g., in the manner discussed above), and identifying the selected lead body as the lead body corresponding to the matching one of the reference electrical fingerprints.

(20) In accordance with a third aspect of the present inventions, an implantable neurostimulation kit is provided. The kit comprises a neurostimulator (e.g., an implantable pulse generator), and a plurality of elongated lead bodies configured for being coupled to the neurostimulator. Each of the lead bodies has a plurality of proximal contacts and a plurality of distal electrodes respectively electrically coupled to the proximal contacts. The in-line connectivity between the electrodes and proximal contacts carried by the different lead bodies differs from each other. By way of non-limiting example, the kit may further comprise at least one spinal cord stimulation lead having the plurality of lead bodies, a surgical paddle lead having the plurality of lead bodies and/or a surgical paddle having one of the plurality of lead bodies, and a percutaneous lead having another of the plurality of lead bodies.

(21) Other and further aspects and features of the invention will be evident from reading the following detailed description of the preferred embodiments, which are intended to illustrate, not limit, the invention.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The drawings illustrate the design and utility of preferred embodiments of the present invention, in which similar elements are referred to by common reference numerals. In order to better appreciate how the above-recited and other advantages and objects of the present inventions are obtained, a more particular description of the present inventions briefly described above will be rendered by reference to specific embodiments thereof, which are illustrated in the accompanying drawings. Understanding that these drawings depict only typical embodiments of the invention and are not therefore to be considered limiting of its scope, the invention will be described and explained with additional specificity and detail through the use of the accompanying drawings in which:

(2) FIG. 1 is plan view of one embodiment of a spinal cord stimulation (SCS) system arranged in

accordance with the present inventions;

(3) FIG. 2 is a plan view of an implantable pulse generator (IPG) and one embodiment of a surgical paddle stimulation lead used in the SCS system of FIG. 1;

(4) FIG. 3 is a plan view of an implantable pulse generator (IPG) and another embodiment of a percutaneous stimulation lead used in the SCS system of FIG. 1;

(5) FIG. 4 is a plan view of the SCS system of FIG. 1 in use with a patient;

(6) FIG. 5 is a block diagram of the internal components of the IPG of FIG. 1;

(7) FIG. 6 is a plan view of a remote control that can be used in the SCS system of FIG. 1;

(8) FIG. 7 is a block diagram of the internal componentry of the remote control of FIG. 6;

(9) FIG. 8 is a block diagram of the components of a clinician's programmer that can be used in the SCS system of FIG. 1;

(10) FIG. 9 is a graph of an exemplary electrical fingerprint generated by a lead body of the percutaneous lead of FIG. 3;

(11) FIG. 10 is a graph of an exemplary electrical fingerprint generated by a first lead body of the surgical paddle lead of FIG. 2; and

(12) FIG. 11 is a graph of an exemplary electrical fingerprint generated by a second lead body of the surgical paddle lead of FIG. 2.

DETAILED DESCRIPTION OF THE EMBODIMENTS

(13) The description that follows relates to a spinal cord stimulation (SCS) system. However, it is to be understood that while the invention lends itself well to applications in SCS, the invention, in its broadest aspects, may not be so limited. Rather, the invention may be used with any type of implantable electrical circuitry used to stimulate tissue. For example, the present invention may be used as part of a pacemaker, a defibrillator, a cochlear stimulator, a retinal stimulator, a stimulator configured to produce coordinated limb movement, a cortical stimulator, a deep brain stimulator, peripheral nerve stimulator, microstimulator, or in any other neural stimulator configured to treat urinary incontinence, sleep apnea, shoulder subluxation, headache, etc.

(14) Turning first to FIG. 1, an exemplary SCS system 10 generally comprises at least one implantable stimulation lead 12 (e.g., surgical paddle lead 12(1) and/or percutaneous lead 12(2)), an implantable pulse generator (IPG) 14, an external remote control RC 16, a Clinician's Programmer (CP) 18, an External Trial Stimulator (ETS) 20, and an external charger 22.

(15) The IPG 14 is physically connected via one or more lead extensions 24 to the stimulation lead(s) 12, which carries a plurality of electrodes 26 (electrodes 26(1)-(3)) arranged in an array. In the illustrated embodiment, one of the stimulation leads that can be incorporated into the SCS system 10 is a surgical paddle lead 12(1) that carries electrodes 26(1)/(2), and another one of the stimulation leads that can be incorporated into the SCS system 10 is a percutaneous lead 12(2) that carries electrodes 26(3). Although only one surgical paddle lead 12(1) and only one percutaneous lead 12(2) are shown, multiple surgical paddle leads 12(1) and/or percutaneous leads 12(2) can be used with the SCS system 10. As will also be described in further detail below, the IPG 14 includes pulse generation circuitry that delivers electrical stimulation energy in the form of a pulsed electrical waveform (i.e., a temporal series of electrical pulses) to the electrode array 26 in accordance with a set of stimulation parameters.

(16) The IPG 14 and stimulation leads 12 can be provided as an implantable neurostimulation kit, along with, e.g., a hollow needle, a stylet, a tunneling tool, and a tunneling straw. Further details discussing implantable kits are disclosed in U.S. Application Ser. No. 61/030,506, entitled "Temporary Neurostimulation Lead Identification Device."

(17) The ETS 20 may also be physically connected via the percutaneous lead extensions 28 and external cable 30 to the stimulation lead 12. The ETS 20, which has similar pulse generation circuitry as the IPG 14, also delivers electrical stimulation energy in the form of a pulse electrical waveform to the electrode array 26 in accordance with a set of stimulation parameters. The major difference between the ETS 20 and the IPG 14 is that the ETS 20 is a non-implantable device that

is used on a trial basis after the stimulation lead **12** has been implanted and prior to implantation of the IPG **14**, to test the responsiveness of the stimulation that is to be provided. Further details of an exemplary ETS are described in U.S. Pat. No. 6,895,280, which is expressly incorporated herein by reference.

(18) The RC **16** may be used to telemetrically control the ETS **20** via a bi-directional RF communications link **32**. Once the IPG **14** and stimulation lead **12** is implanted, the RC **16** may be used to telemetrically control the IPG **14** via a bi-directional RF communications link **34**. Such control allows the IPG **14** to be turned on or off and to be programmed with different stimulation programs after implantation. Once the IPG **14** has been programmed, and its power source has been charged or otherwise replenished, the IPG **14** may function as programmed without the RC **16** being present.

(19) The CP **18** provides clinician detailed stimulation parameters for programming the IPG **14** and ETS **20** in the operating room and in follow-up sessions. The CP **18** may perform this function by indirectly communicating with the IPG **14** or ETS **20**, through the RC **16**, via an IR communications link **36**. Alternatively, the CP **18** may directly communicate with the IPG **14** or ETS **20** via an RF communications link (not shown).

(20) The external charger **22** is a portable device used to transcutaneously charge the IPG **14** via an inductive link **38**. For purposes of brevity, the details of the external charger **22** will not be described herein. Details of exemplary embodiments of external chargers are disclosed in U.S. Pat. No. 6,895,280, which has been previously incorporated herein by reference. Once the IPG **14** has been programmed, and its power source has been charged by the external charger **22** or otherwise replenished, the IPG **14** may function as programmed without the RC **16** or CP **18** being present.

(21) As briefly discussed above, one of the stimulation leads may be a surgical paddle lead **12(1)**. To this end, and with reference to FIG. 2, the surgical paddle lead **12(1)** comprises a paddle-shaped membrane **40**, and two elongated lead bodies **42** (a first lead body **42(1)** and a second lead body **42(2)**) extending from the paddle-shaped membrane **40**. Each of the lead bodies **42** has a proximal end **44** and a distal end **46**. Each lead body **42** may, e.g., have a diameter within the range of 0.03 inches to 0.07 inches and a length within the range of 30 cm to 90 cm for spinal cord stimulation applications. Each lead body **42** may be composed of a suitable electrically insulative material, such as, a polymer (e.g., polyurethane or silicone), and may be extruded from as a unibody construction. The paddle-shaped membrane **40** is composed of an electrically insulative material, such as silicone.

(22) The surgical paddle lead **12(1)** further comprises proximal contacts **48** (proximal contacts **48(1)** and proximal contacts **48(2)**) mounted to the proximal ends **44** of the lead bodies **42** and the plurality of electrodes **26** mounted on one side of the paddle-shaped membrane **40** in a two-dimensional arrangement. In the illustrated embodiment, proximal contacts **48(1)** are mounted to the proximal end **44** of the first lead body **42(1)**, proximal contacts **48(2)** are mounted to the proximal end **44** of the second lead body **42(2)**, and the electrodes **26** are arranged as a first column of electrodes **26(1)** and a second column of electrodes **26(2)** on the membrane **40**. As shown, the electrodes **26** are labeled as electrodes E1-E8 for each column, and the proximal contacts **48** are labeled as proximal contacts PC1-PC8 for each lead body **42**. Although the stimulation lead **12(1)** is shown as having sixteen electrodes **26** (and thus, eight corresponding proximal contacts on each lead body **36**), the number of electrodes may be any number suitable for the application in which the surgical paddle lead **12(1)** is intended to be used (e.g., two, four, eight, etc.).

(23) Each of the electrodes **26** takes the form of a disk composed of an electrically conductive, non-corrosive, material, such as, e.g., platinum, titanium, stainless steel, or alloys thereof. Each of the proximal contacts **48** takes the form of a cylindrical ring element composed of an electrically conductive, biocompatible, non-corrosive, material, such as, e.g., platinum, titanium, stainless steel, or alloys thereof.

(24) The surgical paddle lead **12(1)** also includes a plurality of electrical conductors (not shown)

extending through individual lumens (not shown) within each lead body **42** and connected between the respective proximal contacts **48** and electrodes **26** using suitable means, such as welding, thereby electrically coupling the proximal contacts **48(1)** on the first lead body **42(1)** to the first column of electrodes **26(1)** and the proximal contacts **48(2)** on the second lead body **42(2)** to the second column of electrodes **26(2)**.

(25) Further details regarding the construction and method of manufacture of paddle leads are disclosed in U.S. patent application Ser. No. 11/319,291, entitled "Stimulator Leads and Methods for Lead Fabrication," the disclosure of which is expressly incorporated herein by reference.

(26) Significantly, as will be described in further detail below, the in-line connectivity between the proximal contacts **48(1)** located on the first lead body **42(1)** and the first column of electrodes **26(1)**, and the in-line connectivity between the proximal contacts **48(2)** on the second lead body **42(2)** and the second column of electrodes **26(2)** differ from each other. That is, relative to the same physical ordering between the proximal contacts and electrodes, the order in which the proximal contacts **48(1)** of the first lead body **42(1)** is electrically connected to the first column of electrodes **26(1)** differs from the order in which the proximal contacts **48(2)** of the second lead body **42(2)** is electrically connected to the second column of electrodes **26(2)**. As will be discussed in further detail below, the different in-line connectivities of the lead bodies **42** allows them to be electrically distinguished from each other, and thereby identified.

(27) As briefly discussed above, another of the stimulation leads may be a percutaneous lead **12(3)**. To this end, and with reference to FIG. 3, the percutaneous lead **12(3)** comprises an elongated lead body **42(3)** having a proximal end **44** and a distal end **46**. The lead body **42(3)** may, e.g., have a diameter within the range of 0.03 inches to 0.07 inches and a length within the range of 30 cm to 90 cm for spinal cord stimulation applications. The lead body **42(3)** may be composed of a suitable electrically insulative material, such as, a polymer (e.g., polyurethane or silicone), and may be extruded from as a unibody construction.

(28) The percutaneous lead **12(2)** further comprises a plurality of proximal contacts **48(3)** mounted to the proximal end **44** of the lead body **48(3)** and the plurality of in-line electrodes **26(3)** mounted to the distal end **46** of the lead body **48(3)**. As shown, the electrodes **26(3)** are labeled as electrodes **E1-E8**, and the proximal contacts **48(3)** are labeled as proximal contacts **PC1-PC8**. Although the percutaneous lead **12(2)** is shown as having eight electrodes **26(3)** (and thus, eight corresponding proximal contacts **48(3)**), the number of electrodes may be any number suitable for the application in which the percutaneous lead **12(2)** is intended to be used (e.g., two, four, sixteen, etc.). Each of the electrodes **26** and proximal contacts **48** takes the form of a cylindrical ring element composed of an electrically conductive, biocompatible, non-corrosive, material, such as, e.g., platinum, titanium, stainless steel, or alloys thereof, which is circumferentially disposed about the lead bodies.

(29) The percutaneous lead **12(2)** also includes a plurality of electrical conductors (not shown) extending within the lead body **42(3)** and connected between the respective proximal contacts **48(3)** and electrodes **26(3)** using suitable means, such as welding, thereby electrically coupling the proximal contacts **48(3)** with the distally-located electrodes **26(3)**. The percutaneous lead **12(2)** further includes a central lumen (not shown) that may be used to accept an insertion stylet (described in further detail below) to facilitate lead implantation.

(30) Further details describing the construction and method of manufacturing percutaneous stimulation leads are disclosed in U.S. patent application Ser. No. 11/689,918, entitled "Lead Assembly and Method of Making Same," and U.S. patent application Ser. No. 11/565,547, entitled "Cylindrical Multi-Contact Electrode Lead for Neural Stimulation and Method of Making Same," the disclosures of which are expressly incorporated herein by reference.

(31) Significantly, as will be described in further detail below, the in-line connectivity between the proximal contacts **48(3)** and electrodes **26(3)** of the percutaneous lead **12(2)**, and the in-line connectivities between the proximal contacts **48(1)/(2)** and electrodes **26(1)/(2)** of the surgical

paddle lead **12(1)** differ from each other. That is, relative to the same physical ordering between the proximal contacts and electrodes, the order in which the proximal contacts **48(3)** of the lead body **42(3)** is electrically connected to the electrodes **26(3)** differs from the order in which the proximal contacts **48(1)** of the first lead body **42(1)** of the surgical paddle lead **12(1)** is electrically connected to the first column of electrodes **26(1)**, and further differs from the order in which the proximal contacts **48(2)** of the second lead body **42(2)** of the surgical paddle lead **12(1)**. As will be discussed in further detail below, the different in-line connectivities of the lead bodies **42** allows them to be electrically distinguished from each other, and thereby identified.

(32) Referring to either of FIG. 2 or 3, the IPG **14** comprises an outer case **50** or housing the electronic and other components (described in further detail below). The outer case **50** is composed of an electrically conductive, biocompatible material, such as titanium, and forms a hermetically sealed compartment wherein the internal electronics are protected from the body tissue and fluids. In some cases, the outer case **50** serves as an electrode. The IPG **14** further comprises a connector **52** in which the proximal ends **44** of the lead bodies **42** of the stimulation leads **12** can mate in a manner that electrically couples the electrodes **26** to the electronics contained within the outer case **50**. To this end, the connector **52** includes a pair of ports **54** (only one shown in phantom) for receiving the proximal ends **44** of the lead bodies **42(1)/(2)** of the surgical paddle lead **12(1)** or the proximal end **44** of the lead body **42(3)** of the percutaneous lead **12(2)** (or proximal ends **44** of the lead bodies **42(3)** in the case where two percutaneous leads **12(2)** are used). In the case where the lead extensions **24** are used, the ports **54** may instead receive the proximal ends of such lead extensions **24**.

(33) As will be described in further detail below, the IPG **14** includes pulse generation circuitry that provides electrical stimulation energy to the electrodes **26** in accordance with a set of parameters. Such parameters may comprise electrode combinations, which define the electrodes that are activated as anodes (positive), cathodes (negative), and turned off (zero), and electrical pulse parameters, which define the pulse amplitude (measured in milliamps or volts depending on whether the IPG **14** supplies constant current or constant voltage to the electrodes), pulse duration (measured in microseconds), and pulse rate (measured in pulses per second).

(34) With respect to the pulse patterns provided during operation of the SCS system **10**, electrodes that are selected to transmit or receive electrical energy are referred to herein as “activated,” while electrodes that are not selected to transmit or receive electrical energy are referred to herein as “non-activated.” Electrical energy delivery will occur between two (or more) electrodes, one of which may be the IPG case **50**, so that the electrical current has a path from the energy source contained within the IPG case **50** to the tissue and a sink path from the tissue to the energy source contained within the case. Electrical energy may be transmitted to the tissue in a monopolar or multipolar (e.g., bipolar, tripolar, etc.) fashion.

(35) Monopolar delivery occurs when a selected one or more of the lead electrodes **26** is activated along with the case **50** of the IPG **14**, so that electrical energy is transmitted between the selected electrode **26** and case **50**. Monopolar delivery may also occur when one or more of the lead electrodes **26** are activated along with a large group of lead electrodes located remotely from the one more lead electrodes **26** so as to create a monopolar effect; that is, electrical energy is conveyed from the one or more lead electrodes **26** in a relatively isotropic manner. Bipolar delivery occurs when two of the lead electrodes **26** are activated as anode and cathode, so that electrical energy is transmitted between the selected electrodes **26**. Tripolar delivery occurs when three of the lead electrodes **26** are activated, two as anodes and the remaining one as a cathode, or two as cathodes and the remaining one as an anode.

(36) Referring to FIG. 4, the stimulation lead **12** (either **12(1)** or **12(2)**) is implanted within the spinal column **56** of a patient **58**. The preferred placement of the stimulation lead **12** is adjacent, i.e., resting upon, the spinal cord area to be stimulated. Due to the lack of space near the location where the stimulation lead **12** exits the spinal column **56**, the IPG **14** is generally implanted in a

surgically-made pocket either in the abdomen or above the buttocks. The IPG **14** may, of course, also be implanted in other locations of the patient's body. The lead extension **24** may be used to facilitate locating the IPG **14** away from the exit point of the stimulation lead **12**. After implantation, the IPG **14** is used to provide the therapeutic stimulation under control of the patient. (37) Turning next to FIG. **5**, the main internal components of the IPG **14** will now be described. The IPG **14** includes analog output circuitry **60** capable of individually generating electrical stimulation pulses via capacitors **61** at the electrodes **26** of specified amplitude under control of control logic **62** over data bus **64**. The duration of the electrical stimulation (i.e., the width of the stimulation pulses), is controlled by the timer logic **66**. The analog output circuitry **60** may either comprise independently controlled current sources for providing stimulation pulses of a specified and known amperage to or from the electrodes **26**, or independently controlled voltage sources for providing stimulation pulses of a specified and known voltage at the electrodes **26**. The operation of this analog output circuitry **60**, including alternative embodiments of suitable output circuitry for performing the same function of generating stimulation pulses of a prescribed amplitude and width, is described more fully in U.S. Pat. Nos. 6,516,227 and 6,993,384, which are expressly incorporated herein by reference.

(38) The IPG **14** further comprises monitoring circuitry **70** for monitoring the status of various nodes or other points **72** throughout the IPG **14**, e.g., power supply voltages, temperature, battery voltage, and the like. Notably, the electrodes **26** fit snugly within the epidural space of the spinal column, and because the tissue is conductive, electrical measurements can be taken between the electrodes **26**. Significantly, the monitoring circuitry **70** is configured for taking such electrical measurements, so that, as will be described in further detail below, the CP **18** can automatically identify the specific lead bodies **42** that are connected to the IPG **14**. In the illustrated embodiment, the electrical measurements taken by the monitoring circuitry **70** for the purpose of identifying the connected lead bodies, are field potentials. The monitoring circuitry **70** may also measure impedance at each electrode **26** in order to determine the coupling efficiency between the respective electrode **26** and the tissue and/or to facilitate fault detection with respect to the connection between the electrodes **26** and the analog output circuitry **60** of the IPG **14**.

(39) Electrical data can be measured using any one of a variety of means. For example, the electrical data measurements can be made on a sampled basis during a portion of the time while the electrical stimulus pulse is being applied to the tissue, or immediately subsequent to stimulation, as described in U.S. patent application Ser. No. 10/364,436, which has previously been incorporated herein by reference. Alternatively, the electrical data measurements can be made independently of the electrical stimulation pulses, such as described in U.S. Pat. Nos. 6,516,227 and 6,993,384, which are expressly incorporated herein by reference.

(40) The IPG **14** further comprises processing circuitry in the form of a microcontroller **74** that controls the control logic **62** over data bus **76**, and obtains status data from the monitoring circuitry **70** via data bus **78**. The IPG **14** additionally controls the timer logic **66**. The IPG **14** further comprises memory **80** and an oscillator and clock circuit **82** coupled to the microcontroller **74**. The microcontroller **74**, in combination with the memory **80** and oscillator and clock circuit **82**, thus comprise a microprocessor system that carries out a program function in accordance with a suitable program stored in the memory **80**. Alternatively, for some applications, the function provided by the microprocessor system may be carried out by a suitable state machine.

(41) Thus, the microcontroller **74** generates the necessary control and status signals, which allow the microcontroller **74** to control the operation of the IPG **14** in accordance with a selected operating program and parameters. In controlling the operation of the IPG **14**, the microcontroller **74** is able to individually generate electrical pulses at the electrodes **26** using the analog output circuitry **60**, in combination with the control logic **62** and timer logic **66**, thereby allowing each electrode **26** to be paired or grouped with other electrodes **26**, including the monopolar case electrode, and to control the polarity, amplitude, rate, and pulse width through which the current

stimulus pulses are provided.

(42) The IPG **14** further comprises an alternating current (AC) receiving coil **84** for receiving programming data (e.g., the operating program and/or stimulation parameters) from the RC **16** in an appropriate modulated carrier signal, and charging and forward telemetry circuitry **86** for demodulating the carrier signal it receives through the AC receiving coil **84** to recover the programming data, which programming data is then stored within the memory **80**, or within other memory elements (not shown) distributed throughout the IPG **14**.

(43) The IPG **14** further comprises back telemetry circuitry **88** and an alternating current (AC) transmission coil **90** for sending informational data (including the field potential and impedance data) sensed through the monitoring circuitry **70** to the RC **16**. The back telemetry features of the IPG **14** also allow its status to be checked. For example, any changes made to the stimulation parameters are confirmed through back telemetry, thereby assuring that such changes have been correctly received and implemented within the IPG **14**. Moreover, upon interrogation by the RC **16**, all programmable settings stored within the IPG **14** may be uploaded to the RC **16**.

(44) The IPG **14** further comprises a rechargeable power source **92** and power circuits **94** for providing the operating power to the IPG **14**. The rechargeable power source **92** may, e.g., comprise a lithium-ion or lithium-ion polymer battery. The rechargeable battery **92** provides an unregulated voltage to the power circuits **94**. The power circuits **94**, in turn, generate the various voltages **96**, some of which are regulated and some of which are not, as needed by the various circuits located within the IPG **14**. The rechargeable power source **92** is recharged using rectified AC power (or DC power converted from AC power through other means, e.g., efficient AC-to-DC converter circuits, also known as “inverter circuits”) received by the AC receiving coil **84**. To recharge the power source **92**, an external charger (not shown), which generates the AC magnetic field, is placed against, or otherwise adjacent, to the patient's skin over the implanted IPG **14**. The AC magnetic field emitted by the external charger induces AC currents in the AC receiving coil **84**. The charging and forward telemetry circuitry **86** rectifies the AC current to produce DC current, which is used to charge the power source **92**. While the AC receiving coil **84** is described as being used for both wirelessly receiving communications (e.g., programming and control data) and charging energy from the external device, it should be appreciated that the AC receiving coil **84** can be arranged as a dedicated charging coil, while another coil, such as coil **90**, can be used for bi-directional telemetry.

(45) It should be noted that the diagram of FIG. **5** is functional only, and is not intended to be limiting. Those of skill in the art, given the descriptions presented herein, should be able to readily fashion numerous types of IPG circuits, or equivalent circuits, that carry out the functions indicated and described. It should be noted that rather than an IPG for the neurostimulator, the SCS system **10** may alternatively utilize an implantable receiver-stimulator (not shown) connected to the stimulation lead **12**. In this case, the power source, e.g., a battery, for powering the implanted receiver, as well as control circuitry to command the receiver-stimulator, will be contained in an external controller inductively coupled to the receiver-stimulator via an electromagnetic link. Data/power signals are transcutaneously coupled from a cable-connected transmission coil placed over the implanted receiver-stimulator. The implanted receiver-stimulator receives the signal and generates the stimulation in accordance with the control signals.

(46) Referring now to FIG. **6**, one exemplary embodiment of an RC **16** will now be described. As previously discussed, the RC **16** is capable of communicating with the IPG **14**, CP **18**, or ETS **20**. The RC **16** comprises a casing **100**, which houses internal componentry (including a printed circuit board (PCB)), and a lighted display screen **102** and button pad **104** carried by the exterior of the casing **100**. In the illustrated embodiment, the display screen **102** is a lighted flat panel display screen, and the button pad **104** comprises a membrane switch with metal domes positioned over a flex circuit, and a keypad connector connected directly to a PCB. In an optional embodiment, the display screen **102** has touchscreen capabilities. The button pad **104** includes a multitude of buttons

106, 108, 110, and 112, which allow the IPG **14** to be turned ON and OFF, provide for the adjustment or setting of stimulation parameters within the IPG **14**, and provide for selection between screens.

(47) In the illustrated embodiment, the button **106** serves as an ON/OFF button that can be actuated to turn the IPG **14** ON and OFF. The button **108** serves as a select button that allows the RC **106** to switch between screen displays and/or parameters. The buttons **110** and **112** serve as up/down buttons that can be actuated to increment or decrement any of stimulation parameters of the pulse generated by the IPG **14**, including pulse amplitude, pulse width, and pulse rate.

(48) Referring to FIG. 7, the internal components of an exemplary RC **16** will now be described. The RC **16** generally includes a processor **114** (e.g., a microcontroller), memory **116** that stores an operating program for execution by the processor **114**, and telemetry circuitry **118** for transmitting control data (including stimulation parameters and requests to provide status information) to the IPG **14** (or ETS **20**) and receiving status information (including the measured electrical data) from the IPG **14** (or ETS **20**) via link **34** (or link **32**) (shown in FIG. 1), as well as receiving the control data from the CP **18** and transmitting the status data to the CP **18** via link **36** (shown in FIG. 1). The RC **16** further includes input/output circuitry **120** for receiving stimulation control signals from the button pad **104** and transmitting status information to the display screen **102** (shown in FIG. 6). Further details of the functionality and internal componentry of the RC **16** are disclosed in U.S. Pat. No. 6,895,280, which has previously been incorporated herein by reference.

(49) As briefly discussed above, the CP **18** greatly simplifies the programming of multiple electrode combinations, allowing the physician or clinician to readily determine the desired stimulation parameters to be programmed into the IPG **14**, as well as the RC **16**. Thus, modification of the stimulation parameters in the programmable memory of the IPG **14** after implantation is performed by a clinician using the CP **18**, which can directly communicate with the IPG **14** or indirectly communicate with the IPG **14** via the RC **16**. That is, the CP **18** can be used by the physician or clinician to modify operating parameters of the electrode array **26** near the spinal cord.

(50) As shown in FIG. 4, the overall appearance of the CP **18** is that of a laptop personal computer (PC), and in fact, may be implemented using a PC that has been appropriately configured to include a directional-programming device and programmed to perform the functions described herein. Thus, the programming methodologies can be performed by executing software instructions contained within the CP **18**. Alternatively, such programming methodologies can be performed using firmware or hardware. In any event, the CP **18** may actively control the characteristics of the electrical stimulation generated by the IPG **14** (or ETS **20**) to allow the optimum stimulation parameters to be determined based on patient feedback and for subsequently programming the IPG **14** (or ETS **20**) with the optimum stimulation parameters.

(51) To allow the clinician to perform these functions, the CP **18** includes a mouse **121**, a keyboard **122**, and a programming display screen **124** housed in a case **126**. It is to be understood that in addition to, or in lieu of, the mouse **121**, other directional programming devices may be used, such as a joystick, or directional keys included as part of the keys associated with the keyboard **122**. As shown in FIG. 8, the CP **18** generally includes a processor **128** (e.g., a central processor unit (CPU)) and memory **130** that stores a stimulation programming package **132**, which can be executed by the processor **128** to allow a clinician to program the IPG **14** (or ETS **20**) and RC **16**. The CP **18** further includes telemetry circuitry **134** for downloading stimulation parameters to the RC **16** and uploading stimulation parameters already stored in the memory **116** of the RC **16** via link **36** (shown in FIG. 1). The telemetry circuitry **134** is also configured for transmitting the control data (including stimulation parameters and requests to provide status information) to the IPG **14** (or ETS **20**) and receiving status information (including the measured electrical data) from the IPG **14** (or ETS **20**) indirectly via the RC **16**.

(52) Referring further to FIGS. 2 and 3, the CP **18** is configured for automatically identifying the lead body **42** of the stimulation lead **12** that is coupled to the IPG **14** based on an electrical

fingerprint created by the lead body **42**. In particular, the CP **18** is configured for operating the IPG **14** (i.e., by transmitting a request for status information to the IPG **14** via the RC **16**) to measure an electrical fingerprint at the proximal contacts **48** of the lead body **42**. Once the lead body **42** is identified, the CP **18** is configured for remapping the outputs of the analog output circuitry **60** (shown in FIG. 5) to the proper electrodes **26** of the stimulation lead **12**. Once the lead body **42** is identified and the outputs of the analog output circuitry **60** are mapped to the electrodes **26**, the CP **18** can then generate stimulation parameters for use by the IPG **14** based on the identified lead body **42**. Thus, the user may insert the lead body **42** into any port **54** of the connector **52** of the IPG **14** without concern that the incorrect connector port is being used.

(53) In the illustrated embodiment, the electrical fingerprint can be correlated to the distances that each electrode **26** is from the other electrodes **26** in the electrode column associated with the distal end of the lead **12**.

(54) To this end, the electrical fingerprint is generated and measured by first passing electrical current through each proximal contact **48** associated with the lead body **42** to be identified. The electrical current may be passed in a monopolar arrangement; for example, between the respective proximal contact **48** and the IPG case **50**, or in a multipolar arrangement; for example, between the respective proximal contact **48** and one or more other proximal contacts **48** carried by the lead body **42**. In response to the passing of electrical current through each respective proximal contact **48**, the field potential is measured at each of the other proximal contacts **48** to create field potential profiles for each proximal contact **48**.

(55) Thus, each measured field potential is correlated to a distance between the electrode **26** corresponding to the proximal contact **48** through which the electrical current passes and the electrode **26** corresponding to the proximal contact **48** at which the field potential is measured. From this, field potential profiles for each proximal contact **48** can be created, a composite of which can be used to generate a fingerprint for the lead body **42** on which the proximal contacts **48** and corresponding electrodes **26** are carried. Further details regarding field potential measurement techniques are disclosed in U.S. Pat. No. 6,993,384, which is expressly incorporated herein by reference.

(56) The uniqueness of the electrical fingerprints is provided by the different in-line connectivities between the proximal contacts **48** and electrodes **26** associated with the respective lead bodies **42**. That is, for the same arrangement of proximal contacts **48**, changing the in-line connectivities will change the distances between the electrodes **26** corresponding to the proximal contacts **48**, thereby changing the field potential profiles for each proximal contact **48**, and the resulting electrical fingerprint for the lead body **42**. Notably, without changing the in-line connectivities, the two columns of electrodes **26(1)/(2)** associated with the respective lead bodies **42(1)/(2)** of the surgical paddle lead **12(1)**, as well as the electrodes **26(3)** associated with the lead body **42(3)** of the percutaneous lead **12(2)**, which may have the same rostro-caudal electrode spacing, will create very similar electrical fingerprints. Thus, the electrical fingerprints created by the lead bodies **42(1)/(2)** of the surgical paddle lead **12(1)** and by the lead body **42(3)** of the percutaneous lead **12(2)** would not be distinguishable from each other if the connectivities were the same.

(57) For example, with respect to the percutaneous lead **12(2)**, the proximal contacts PC1-PC8 may be consecutively connected to the electrodes E1-E8 in a conventional manner. That is, proximal contact PC1 is connected to electrode E1, proximal contact PC2 is connected to electrode E2, proximal contact PC3 is connected to electrode E3, proximal contact PC4 is connected to electrode E4, proximal contact PC5 is connected to electrode E5, proximal contact PC6 is connected to electrode E6, proximal contact PC7 is connected to electrode E7, and proximal contact PC8 is connected to electrode E8.

(58) As shown in FIG. 9, the resulting electrical fingerprint of the percutaneous lead **12(2)** comprises a composite of field potential profiles for respective proximal contacts PC1-PC8. In this case, the field potential profile for each of proximal contacts PC1-PC8 is created by applying

electrical current to each respective proximal contact in a monopolar fashion and measuring the field potential at the other proximal contacts. That is, field potentials are measured at proximal contacts PC2-PC8 while electrical current is applied to proximal contact PC1; field potentials are measured at proximal contacts PC1 and PC3-PC8 while electrical current is applied to proximal contact PC2; field potentials are measured at proximal contacts PC1-PC2 and PC4-PC8 while electrical current is applied to proximal contact PC3, etc.

(59) The spacing of the electrodes respectively connected to proximal contacts PC2-PC8 determines the field potential profile of proximal contact PC1. In particular, the field potential profile for proximal contact PC1 has the greatest value at proximal contact PC2, since corresponding electrode E1 is spatially closest to corresponding electrode E2. The field potential profile decreases from this greatest value in the order of proximal contacts PC3-PC8, since the spacing between corresponding electrode E1 and corresponding electrodes E3-E8 incrementally increases in that order.

(60) The spacing of the electrodes respectively connected to proximal contacts PC1 and PC3-PC8 determines the field potential profile of proximal contact PC2. In particular, the field potential profile for proximal contact PC2 has the greatest value at proximal contacts PC1 and PC3, since corresponding electrode E2 is spatially closest to corresponding electrodes E1 and E3. The field potential profile decreases from this greatest value in the order of proximal contacts PC4-PC8, since the spacing between corresponding electrode E2 and corresponding electrodes E4-E8 incrementally increases in that order.

(61) The spacing of the electrodes respectively connected to proximal contacts PC1-PC2 and PC4-PC8 determines the field potential profile of proximal contact PC3. In particular, the field potential profile for proximal contact PC3 has the greatest value at proximal contacts PC2 and PC4, since corresponding electrode E3 is spatially closest to corresponding electrodes E2 and E4. The field potential profile increases to this greatest value from proximal contact PC1, and then decreases from this greatest value in the order of proximal contacts PC5-PC8, since the spacing between corresponding electrode E3 and corresponding electrodes E5-E8 incrementally increases in that order.

(62) The spacing of the electrodes respectively connected to proximal contacts PC1-PC3 and PC5-PC8 determines the field potential profile of proximal contact PC4. In particular, the field potential profile for proximal contact PC4 has the greatest value at proximal contacts PC3 and PC5, since corresponding electrode E4 is spatially closest to corresponding electrodes E3 and E5. The field potential profile increases to this greatest value in the order of proximal contacts PC1-PC2, since the spacing between corresponding electrode E4 and corresponding electrodes E1-E2 incrementally decreases in that order, and then decreases from this greatest value in the order of proximal contacts PC6-PC8, since the spacing between corresponding electrode E4 and corresponding electrodes E6-E8 incrementally increases in that order.

(63) The spacing of the electrodes respectively connected to proximal contacts PC1-PC4 and PC6-PC8 determines the field potential profile of proximal contact PC5. In particular, the field potential profile for proximal contact PC5 has the greatest value at proximal contacts PC4 and PC6, since corresponding electrode E5 is spatially closest to corresponding electrodes E4 and E6. The field potential profile increases to this greatest value in the order of proximal contacts PC1-PC3, since the spacing between corresponding electrode E5 and corresponding electrodes E1-E3 incrementally decreases in that order, and then decreases from this greatest value in the order of proximal contacts PC7-PC8, since the spacing between corresponding electrode E5 and corresponding electrodes E7-E8 incrementally increases in that order.

(64) The spacing of the electrodes respectively connected to proximal contacts PC1-PC5 and PC7-PC8 determines the field potential profile of proximal contact PC6. In particular, the field potential profile for proximal contact PC6 has the greatest value at proximal contacts PC5 and PC7, since corresponding electrode E6 is spatially closest to corresponding electrodes E5 and E7. The field

potential profile increases to this greatest value in the order of proximal contacts PC1-PC4, since the spacing between corresponding electrode E6 and corresponding electrodes E1-E4 incrementally decreases in that order, and then decreases from this greatest value to proximal contact PC8.

(65) The spacing of the electrodes respectively connected to proximal contacts PC1-PC6 and PC8 determines the field potential profile of proximal contact PC7. In particular, the field potential profile for the proximal contact PC7 has the greatest value at proximal contacts PC6 and PC8, since corresponding electrode E7 is spatially closest to corresponding electrodes E6 and E8. The field potential profile increases to this greatest value in the order of proximal contacts PC1-PC5, since the spacing between corresponding electrode E7 and corresponding electrodes E1-E5 incrementally decreases in that order.

(66) The spacing of the electrodes respectively connected to proximal contacts PC1-PC7 determines the field potential profile of proximal contact PC8. In particular, the field potential profile for proximal contact PC8 has the greatest value at proximal contact PC7, since corresponding electrode E8 is spatially closest to corresponding electrode E7. The field potential profile increases to this greatest value in the order of proximal contacts PC1-PC6, since the spacing between corresponding electrode E8 and corresponding electrodes E1-E6 incrementally decreases in that order.

(67) With respect to the first lead body 42(1) of the surgical paddle lead 12(1), proximal contacts PC1-PC8 may be connected to electrodes E1-E8 in an unconventional manner. For example, in the illustrated embodiment, proximal contact PC1 is connected to electrode E2, proximal contact PC2 is connected to electrode E5, proximal contact PC3 is connected to electrode E8, proximal contact PC4 is connected to electrode E1, proximal contact PC5 is connected to electrode E4, proximal contact PC6 is connected to electrode E7, proximal contact PC7 is connected to electrode E3, and proximal contact PC8 is connected to electrode E6.

(68) As shown in FIG. 10, the resulting electrical fingerprint of the first lead body 42(1) of the surgical paddle lead 12(1) comprises a composite of field potential profiles for the respective proximal contacts PC1-PC8. In the same manner described above with respect to the percutaneous lead 12(2), the field potential profile for each of proximal contacts PC1-PC8 is created by applying electrical current to each respective proximal contact in a monopolar fashion and measuring the field potential at the other proximal contacts. In addition, the field potential profile for each proximal contact is determined by the spacing of the electrodes respectively connected to the remaining proximal contacts.

(69) Thus, the field potential profile for proximal contact PC1 has the greatest value at proximal contacts PC4 and PC7, since electrode E2, which is connected to proximal contact PC1, is spatially closest to electrodes E1 and E3, which are respectively connected to proximal contacts PC4 and PC7. The field potential profile for proximal contact PC2 has the greatest value at proximal contacts PC5 and PC8, since electrode E5, which is connected to proximal contact PC2, is spatially closest to electrodes E4 and E6, which are respectively connected to proximal contacts PC5 and PC8. The field potential profile for proximal contact PC3 has the greatest value at proximal contact PC6, since electrode E8, which is connected to proximal contact PC3, is spatially closest to electrode E7, which is connected to proximal contact PC6. The field potential profile for proximal contact PC4 has the greatest value at proximal contact PC1, since electrode E1, which is connected to proximal contact PC4, is spatially closest to electrode E2, which is connected to proximal contact PC1.

(70) The field potential profile for proximal contact PC5 has the greatest value at proximal contacts PC2 and PC7, since electrode E4, which is connected to proximal contact PC5, is spatially closest to electrodes E5 and E3, which are respectively connected to proximal contacts PC2 and PC7. The field potential profile for proximal contact PC6 has the greatest value at proximal contacts PC3 and PC8, since electrode E7, which is connected to proximal contact PC6, is spatially closest to electrodes E8 and E7, which are respectively connected to proximal contacts PC3 and PC8. The

field potential profile for proximal contact PC7 has the greatest value at proximal contacts PC1 and PC5, since electrode E3, which is connected to proximal contact PC7, is spatially closest to electrodes E2 and E4, which are respectively connected to proximal contacts PC1 and PC5. The field potential profile for proximal contact PC8 has the greatest value at proximal contacts PC2 and PC6, since electrode E6, which is connected to proximal contact PC8, is spatially closest to electrodes E5 and E7, which are respectively connected to proximal contacts PC2 and PC6.

(71) Unlike with the percutaneous lead 12(2) where the field potential profiles for each of the proximal contacts PC1-PC8 incrementally increases and decreases from the greatest values, as can be seen in FIG. 10, the field potential profile for each of proximal contacts PC1-PC8 varies erratically from the greatest values, because proximal contacts PC1-PC8 are not consecutively connected to electrodes E1-E8 with respect to the first lead body 42(1) of the surgical paddle lead 12(1).

(72) With respect to the second lead body 42(2) of the surgical paddle lead 12(1), the proximal contacts PC1-PC8 may be connected to electrodes E1-E8 in another unconventional manner. For example, in the illustrated embodiment, proximal contact PC1 is connected to electrode E7, proximal contact PC2 is connected to electrode E6, proximal contact PC3 is connected to electrode E8, proximal contact PC4 is connected to electrode E2, proximal contact PC5 is connected to electrode E4, proximal contact PC6 is connected to electrode E5, proximal contact PC7 is connected to electrode E3, and proximal contact PC8 is connected to electrode E1.

(73) As shown in FIG. 11, the resulting electrical fingerprint of the second lead body 42(2) of the surgical paddle lead 12(1) comprises a composite of field potential profiles for the respective proximal contacts PC1-PC8. In the same manner described above with respect to the percutaneous lead 12(2), the field potential profile for each of proximal contacts PC1-PC8 is created by applying electrical current to each respective proximal contact in a monopolar fashion and measuring the field potential at the other proximal contacts. In addition, the field potential profile for each proximal contact is determined by the spacing of the electrodes respectively connected to the remaining proximal contacts.

(74) Thus, the field potential profile for proximal contact PC1 has the greatest value at proximal contacts PC2 and PC3, since electrode E7, which is connected to proximal contact PC1, is spatially closest to electrodes E6 and E8, which are respectively connected to proximal contacts PC2 and PC3. The field potential profile for proximal contact PC2 has the greatest value at proximal contacts PC1 and PC6, since electrode E6, which is connected to proximal contact PC2, is spatially closest to electrodes E7 and E5, which are respectively connected to proximal contacts PC1 and PC6. The field potential profile for proximal contact PC3 has the greatest value at proximal contact PC1, since electrode E8, which is connected to proximal contact PC3, is spatially closest to electrode E7, which is connected to proximal contact PC1. The field potential profile for proximal contact PC4 has the greatest value at proximal contacts PC7 and PC8, since electrode E2, which is connected to proximal contact PC4, is spatially closest to electrodes E3 and E1, which are respectively connected to proximal contacts PC7 and PC8.

(75) The field potential profile for proximal contact PC5 has the greatest value at proximal contacts PC6 and PC7, since electrode E4, which is connected to proximal contact PC5, is spatially closest to electrodes E5 and E3, which are respectively connected to proximal contacts PC6 and PC7. The field potential profile for proximal contact PC6 has the greatest value at proximal contacts PC2 and PC5, since electrode E5, which is connected to proximal contact PC6, is spatially closest to electrodes E6 and E5, which are respectively connected to proximal contacts PC2 and PC5. The field potential profile for proximal contact PC7 has the greatest value at proximal contacts PC4 and PC5, since electrode E3, which is connected to proximal contact PC7, is spatially closest to electrodes E2 and E4, which are respectively connected to proximal contacts PC4 and PC5. The field potential profile for proximal contact PC8 has the greatest value at proximal contact PC4, since electrode E1, which is connected to proximal contact PC8, is spatially closest to electrode E2,

which is connected to proximal contact PC4.

(76) Again, because proximal contacts PC1-PC8 are not consecutively connected to electrodes E1-E8 with respect to the second lead body 42(2) of the surgical paddle lead 12(1), the field potential profile for each of proximal contacts PC1-PC8 varies erratically from the greatest values, as can be seen in FIG. 11.

(77) The CP 18 automatically identifies the specific lead body 42 connected to the IPG 14 (or ETS 20) based on the electrical fingerprints measured by the IPG 14 (or ETS 20). To this end, the memory 130 stores a plurality of reference electrical fingerprints corresponding to the different lead bodies 42. The reference electrical fingerprints can be generated from expected electrical fingerprints computed from point sources of the electrodes. Alternatively, the reference electrical fingerprints can be generated from actual measurements taken from the proximal contacts of leads. In any event, the processor 128 is configured for comparing the electrical fingerprint measured by the IPG 14 to each of the reference electrical fingerprints stored in memory 130, determining a match between the measured electrical fingerprint and one of the reference electrical fingerprints, and identifying the lead body 42 connected to the IPG 14 as the lead body corresponding to the matching one of the reference electrical fingerprints.

(78) In the illustrated embodiment, the processor 128 is configured for computationally comparing data points in the measured electrical fingerprint to data points in each of the reference electrical fingerprints, and determining a match between the measured electrical fingerprint and the reference electrical fingerprint based on the comparison function.

(79) For example, one comparison function that can be used is a correlation coefficient function, such as a Pearson Correlation Coefficient function, which can be expressed as the following equation:

$$r = \frac{\sum_{i=1}^n (MEAS_{sub.i} - M_{sub}) (REF_{sub.i} - M_{sub})}{\sqrt{\sum_{i=1}^n (MEAS_{sub.i} - M_{sub})^2 \sum_{i=1}^n (REF_{sub.i} - M_{sub})^2}}$$

where r is the coefficient, MEAS represents the data set of the electrical fingerprint measured by the IPG 14, REF represents the data set of the reference electrical fingerprint to which the data of the measured electrical fingerprint is currently be compared, M represents the mean of the data set (either measured or reference), and i represents a single element of the data set (either measured or reference).

(80) Advantageously, the correlation coefficient is not sensitive to magnitude scaling, and ranges from -1 (perfect inverse correlation) to 1 (perfect correlation). Thus, the correlation coefficient would be close to 1 for a match between the measured electrical fingerprint and the reference electrical fingerprint and less than a threshold value (e.g., 0.9) for a mismatch between the measured electrical fingerprint and the reference electrical fingerprint.

(81) The expected correlation coefficient generated by performing a Pearson Correlation Coefficient function on the electrical fingerprints illustrated in FIGS. 9 and 10 is -0.35 ; the expected correlation coefficient generated by performing a Pearson Correlation Coefficient function on the electrical fingerprints illustrated in FIGS. 9 and 11 is 0.25 ; and the expected correlation coefficient generated by performing a Pearson Correlation Coefficient function on the electrical fingerprints illustrated in FIGS. 10 and 11 is 0.14 . Thus, the auto-identification feature of the processor 128 is robust against mismatching a connected lead body 42 to an identified lead body.

(82) Another comparison function that can be used is a least squares based function, and in particular, a sum of squared differences function, which can be expressed as the following equation:

$SSD = \sum_{i=1}^n (MEAS_{sub.i} - REF_{sub.i})^2$, where SSD is the sum of squared difference, and MEAS, REF, and i have been defined above. The SSD function measures the difference between the data set of the measured electrical fingerprint and the data set of the reference electrical fingerprint. With this function, the reference electrical fingerprint that results in the minimum sum

of squared difference is the one whose corresponding lead body is selected as the identified lead body.

(83) Other comparison functions, including cross-correlation functions, wavelet functions, and associated matching measures, may be alternatively used.

(84) It should be noted that the data sets of the electrical fingerprints may be derived from a subset of the electrodes before initially performing the computation function. For example, the auto-identification feature should be robust against open contacts and high impedances. In this case, a smaller number (perhaps one or two) of the proximal contacts **48** exhibiting high impedance values can be ignored. This increased robustness would come at the expense of a reduced number of unique electrical fingerprints that could be encoded into proximal ends of lead bodies.

(85) Notably, a properly seated connection between the proximal contacts **48** of a respective lead body **42** and the connector **52** of the IPG **14** is important for the auto-identify feature to operate properly, since any shift in connector alignment will result in an electrical fingerprint mismatch even if the measured electrical fingerprint is compared to an otherwise matching reference electrical fingerprint. In particular, electrical fingerprints that correspond to various misalignment conditions (conditions where the proximal contacts are misaligned with the connector of the IPG) can be generated. In this case, high impedance measurements, which indicate a misalignment condition, will be retained as part of the electrical fingerprint. The measured electrical fingerprint can then be compared to these reference misalignment electrical fingerprints, so that the CP **18** can automatically identify and correct the misalignment condition.

(86) Also, repeated electrical fingerprint measurements made over time increase the probability of eventually obtaining an incorrect lead body identification. To avoid this, the processor **128** is configured for storing the lead body identifier within the memory **130** when it has first been made and for updating the lead body identifier only when the stimulation lead is replaced and the auto-identify feature is again performed. The auto-identify feature can be performed whenever a programming session is started, but a new lead body identification, if found to be different from the saved lead body identification, would require user confirmation to be applied and overwritten into the memory **130**. This confirmation may also be needed the first time that the lead bodies are connected to the IPG **14** if found to be different from the default consecutive in-line connectivity.

(87) Although particular embodiments of the present inventions have been shown and described, it will be understood that it is not intended to limit the present inventions to the preferred embodiments, and it will be obvious to those skilled in the art that various changes and modifications may be made without departing from the spirit and scope of the present inventions. Thus, the present inventions are intended to cover alternatives, modifications, and equivalents, which may be included within the spirit and scope of the present inventions as defined by the claims.

Claims

1. A method for delivering electrical stimulation to a patient using a stimulator and a plurality of distal electrodes electrically connected to the stimulator through one or more lead bodies each including proximal contacts electrically connected to respective distal electrodes of the plurality of distal electrodes and connected to a connector of the stimulator, the method comprising: measuring an electrical fingerprint of each lead body of the one or more lead bodies, the electrical fingerprint including a set of field potential profiles, the determination including: passing electrical currents through the proximal contacts of the each lead body to result in field potentials each at a contact of the proximal contacts; measuring the field potentials; and creating the set of field potential profiles for the proximal contacts using the field potentials measured for the respective proximal contacts; comparing the measured electrical fingerprint of the each lead body to each reference electrical fingerprint of one or more reference electrical fingerprints of the each lead body; and programming

the stimulator according to an outcome of the comparison.

2. The method of claim 1, wherein comparing the measured electrical fingerprint of the each lead body to the each reference electrical fingerprint of the each lead body comprises comparing a subset of the set of field potential profiles of the measured electrical fingerprint to a respective subset of the set of field potential profiles of the each reference electrical fingerprint.

3. The method of claim 1, further comprising identifying misalignment between the proximal contacts of the each lead body and the connector of the stimulator using an outcome of the comparison.

4. The method of claim 3, further comprising storing the measured electrical fingerprint of the lead body as a reference misalignment electrical fingerprint of the each lead body in response to an identification of the misalignment, the reference misalignment electrical fingerprint being a new reference electrical fingerprint added to the one or more reference electrical fingerprints of the each lead body.

5. The method of claim 4, further comprising measuring the electrical fingerprint of the each lead body for each of various misalignment conditions to store multiple reference misalignment electrical fingerprints of the each lead body.

6. The method of claim 1, further comprising measuring the electrical fingerprint of the each lead body and comparing the measured electrical fingerprint of the each lead body to the each reference electrical fingerprint of the each lead body at start of each programming session for the stimulator.

7. The method of claim 6, further comprising updating a reference electrical fingerprint of one or more reference electrical fingerprints of the each lead body in response to the outcome of the comparison indicating no match between the measured electrical fingerprint of the lead body and any reference electrical fingerprint of the one or more reference electrical fingerprints of the each lead body.

8. The method of claim 7, further comprising storing the updated reference electrical fingerprint as a new reference electrical fingerprint of the each lead body replacing a stored reference electrical fingerprint of the one or more reference electrical fingerprints of the each lead body upon user confirmation.

9. The method of claim 1, wherein comparing the measured electrical fingerprint of the each lead body to the each reference electrical fingerprint of the each lead body comprises determining a correlation coefficient function to compare data points in each profile of the set of field potential profiles of the measured electrical fingerprint of the each lead body to data points in each respective profile of each set of reference field profiles of the each reference electrical fingerprint.

10. The method of claim 1, wherein measuring the electrical fingerprint of the each lead body comprises measuring the electrical fingerprint of each lead body of multiple lead bodies of a surgical paddle lead configured for delivering the spinal cord stimulation.

11. A system for delivering electrical stimulation to a patient using a stimulator and a plurality of distal electrodes configured to be electrically connected to the stimulator through one or more lead bodies each including proximal contacts electrically connected to respective distal electrodes of the plurality of distal electrodes and configured to be connected to a connector of the stimulation, the system comprising: a programmer configured to program the stimulator, the programmer configured to: measure an electrical fingerprint including a set of field potential profiles of each lead body of the one or more lead bodies by programming the stimulator to deliver electrical currents through the proximal contacts of the each lead body to result in field potentials each at a contact of the proximal contacts, measuring the field potentials, and creating the set of field potential profiles for the proximal contacts using the field potentials measured for the respective proximal contacts; compare the measured electrical fingerprint of the each lead body to each reference electrical fingerprint of one or more reference electrical fingerprints of the each lead body; and program the stimulator according to an outcome of the comparison.

12. The system of claim 11, wherein the programmer is configured to compare a subset of the set of

field potential profiles of the measured electrical fingerprint to a respective subset of the set of field potential profiles of the each reference electrical fingerprint.

13. The system of claim 11, wherein the programmer is configured to identify misalignment between the proximal contacts of the each lead body and the connector of the stimulator using the outcome of the comparison.

14. The system of claim 13, wherein the programmer is configured to store the measured electrical fingerprint of the lead body as a reference misalignment electrical fingerprint of the each lead body in response to an identification of the of misalignment, the reference misalignment electrical fingerprint being a new reference electrical fingerprint added to the one or more reference electrical fingerprints of the each lead body.

15. The system of claim 13, wherein the programmer is further configured to identify misalignment between the proximal contacts of the each lead body and the connector of the stimulator using the outcome of the comparison at start of each programming session for the stimulator.

16. The system of claim 11, further comprising the stimulator and a neurostimulation lead including the plurality of distal electrodes and the one or more lead bodies each including the proximal contacts, wherein the stimulator is an implantable stimulator, and the neurostimulation lead is a surgical paddle lead configured for delivering the spinal cord stimulation and including multiple lead bodies.

17. A non-transitory computer-readable storage medium including instructions, which when executed by a system, cause the system to perform a method for delivering electrical stimulation to a patient using a stimulator and a plurality of distal electrodes electrically connected to the stimulator through one or more lead bodies each including proximal contacts electrically connected to respective distal electrodes of the plurality of distal electrodes and connected to a connector of the stimulation, the method comprising: measuring an electrical fingerprint of each lead body of the one or more lead bodies, the electrical fingerprint including a set of field potential profiles, the determination including: passing electrical currents through the proximal contacts of the each lead body to result in field potentials each at a contact of the proximal contacts; measuring the field potentials; and creating the set of field potential profiles for the proximal contacts using the field potentials measured for the respective proximal contacts; comparing the measured electrical fingerprint of the each lead body to each reference electrical fingerprint of one or more reference electrical fingerprints of the each lead body; and determining information for programming the stimulator according to an outcome of the comparison.

18. The non-transitory computer-readable storage medium of claim 17, wherein the method comprises identifying misalignment between the proximal contacts of the each lead body and the connector of the stimulator using an outcome of the comparison.

19. The non-transitory computer-readable storage medium of claim 18, wherein the method further comprises storing the measured electrical fingerprint of the lead body as a reference misalignment electrical fingerprint of the each lead body in response to an identification of the misalignment, the reference misalignment electrical fingerprint being a new reference electrical fingerprint added to the one or more reference electrical fingerprints of the each lead body.

20. The non-transitory computer-readable storage medium of claim 19, wherein the method further comprises measuring the electrical fingerprint of the each lead body for each of various misalignment conditions to store multiple reference misalignment electrical fingerprints of the each lead body.
